7 **(a)** Radioactive decay is both *random* and *spontaneous*. State what is meant by:

(i) random

A random process is one in which it cannot be predicted which nucleus will decay at a particular instant or when any particular nucleus will decay. [1]

(ii) spontaneous

A spontaneous process is one which is not triggered or affected by external factors such as temperature and pressure. [1]

(b) Polonium-193 decays into lead-189 and an alpha particle. The binding energy per nucleon of the particles are given in Table 7.1.

particle	binding energy per nucleon / MeV
$^{193}_{84}Po$	7.774
$^{189}_{82}Pb$	7.826
$^{4}_{2}He$	7.074

Table 7.1

(i) Determine the energy, in MeV, released when a nucleus of polonium-193 decays into a nucleus of lead-189.

energy released = B.E. of Pb-189 + B.E. of helium – B.E. of Po-193

= (189 x 7.826) + (4 x 7.074) – (193 x 7.774) [1]

= 7.028 = 7.03 MeV [1]

energy = ... MeV [2]

(ii) At an instance in time, a block of polonium was found to be emitting energy at a rate of 16.1 W. Calculate the activity of the polonium block.

Energy released per decay in J $= 7.028 \times 10^{6} \times 1.60 \times 10^{-19} = 1.12448 \times 10^{-12}$ J [1]

Power emitted = no. of decays per second x energy per decay

Activity = no. of decays per second = power / energy per decay

$= 16.1 / (1.12448 \times 10^{-12})$

$= 1.4318 \times 10^{13}$

$= 1.43 \times 10^{13}$ Bq [1]

activity = ... Bq [2]

(iii) The half-life of polonium-193 is approximately 420 ms. Calculate the time it will take a pure sample of polonium to decay to 1.0% of its initial amount.

$$\text{Decay constant } \lambda = \frac{\ln 2}{\tau_{1/2}}$$
$$= \frac{\ln 2}{420 \times 10^{-3}}$$
$$= 1.6504 \text{ s}^{-1} \quad [1]$$

$$N = N_0 e^{-\lambda t}$$
$$\text{Tme taken } t = \frac{\ln\left(\frac{N}{N_0}\right)}{-\lambda}$$
$$= \frac{\ln 0.01}{-1.6504}$$
$$= 2.79 \text{ s} \quad [1]$$

time = ... s [2]

[Total: 8]